Project Initialization and Planning Phase

Date	25 June 2026
Team ID	
Project Name	Rising Waters – A Flood Prediction
Maximum Marks	3 Marks

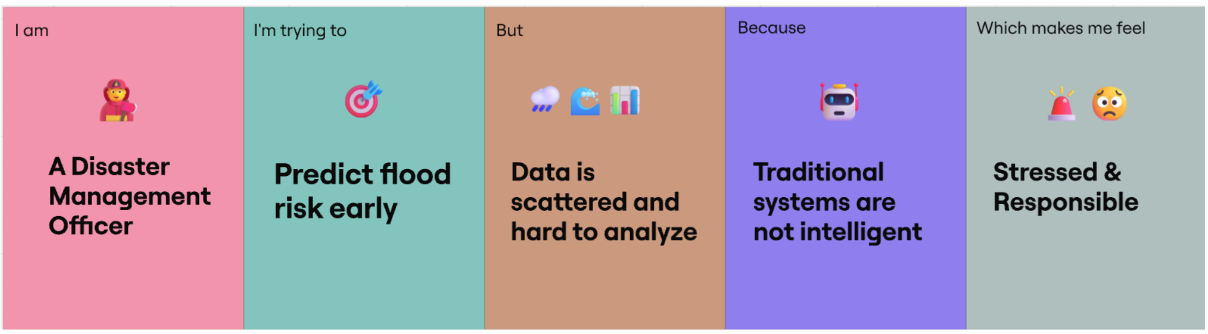
Problem Statement:

Floods are among the most destructive natural disasters in India, causing severe damage to human life, agriculture, transportation, infrastructure, and the economy. Existing flood warning systems often depend on limited rainfall observations or delayed manual assessments, making it difficult for authorities to respond quickly to rapidly changing environmental conditions.

Modern flood prediction requires analysing multiple interconnected factors such as rainfall intensity, river discharge, water level, elevation, soil characteristics, land cover, historical flood occurrences, population density, and flood-control infrastructure. Manually interpreting these diverse parameters is complex, time-consuming, and often results in delayed emergency response.

The objective of this project is to develop an intelligent Machine Learning–based Flood Risk Prediction System capable of analysing multiple environmental and geographical factors simultaneously to estimate flood risk with high accuracy. The proposed system will integrate the trained prediction model into a Flask web application that provides real-time flood risk assessment and decision support for disaster management authorities, local governments, emergency responders, farmers, and citizens. By enabling proactive planning and timely evacuation strategies, the system aims to minimize human casualties, economic losses, and environmental damage caused by floods.

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A District Disaster Management Officer	Identify high-risk flood zones before heavy rainfall causes emergencies	I receive weather, river, and geographical information from different sources	Existing systems do not intelligently combine these factors to provide reliable flood risk predictions	Responsible and under pressure because delayed decisions may cost lives
PS-2	A farmer living near a river basin	Protect my crops, livestock, and family before flooding occurs	I only receive general weather forecasts that do not indicate my area's actual flood risk	Flood occurrence depends on many environmental factors beyond rainfall alone	Uncertain and worried about losing my livelihood